

Lesson 1 · Powers, Factorising and Algebraic Fractions

The one idea: factorise first

Cancelling, combining, solving and rearranging all work on **factors** (things multiplied), never on **terms** (things added).

$$\frac{3x}{5x} = \frac{3}{5} \quad \text{but} \quad \frac{x+3}{x+5} \text{ does not cancel: try } x=1, \text{ you get } \frac{4}{6} \neq \frac{3}{5}.$$

So the first move in almost every question in this lesson is the same: **factorise**, then act on whole factors.

How the paper is marked

- Three questions, each with parts (a), (b), (c), ... Each part is evidence at one level: **u** (Achievement), **r** (Merit) or **t** (Excellence).
- Each question gets a grade score: N1 partial · N2 one u · A3 two u · A4 three u · M5 one r · M6 two r · E7 t with a minor error · E8 t complete.
- The three scores are added (out of 24). The cut scores move a little each year: Achievement starts at about 7–8, Merit at about 13–14, Excellence at about 19–20.
- **Show all working.** A bare answer to a “show that” or “prove” part earns nothing. The formula sheet gives the quadratic formula and the log laws, **not** the index laws or the factorising methods on this handout.

1 Powers and roots (indices)

Law	Example	Law	Example
$x^a \times x^b = x^{a+b}$	$x^4 \times x^6 = x^{10}$	$x^{-n} = \frac{1}{x^n}$	$2^{-3} = \frac{1}{8}$
$\frac{x^a}{x^b} = x^{a-b}$	$\frac{y^3}{y^{-2}} = y^{3-(-2)} = y^5$	$x^{\frac{1}{n}} = \sqrt[n]{x}$	$8^{\frac{1}{3}} = 2$
$(x^a)^b = x^{ab}$	$(a^3)^5 = a^{15}$	$x^{\frac{m}{n}} = (\sqrt[n]{x})^m$	$8^{\frac{2}{3}} = 2^2 = 4$
$(kx)^n = k^n x^n$	$(2x)^3 = 8x^3$	$\sqrt[n]{x^k} = x^{\frac{k}{n}}$	$\sqrt[3]{x^{12}} = x^4$
$x^0 = 1$	$p^0 = 1$		

Where does the minus belong?

A negative index belongs only to the factor it is written on. Moving that factor across the fraction line makes the index positive.

$$3y^{-1} = \frac{3}{y} \quad (3y)^{-1} = \frac{1}{3y} \quad \frac{1}{x^{-2}} = x^2$$

Write it in this order (simplifying a power expression)

1. **Brackets first:** raise **every** factor inside to the power, the number too: $(3x^2)^3 = 27x^6$.
2. **Numbers:** simplify the number part like an ordinary fraction.
3. **Each letter:** add indices when multiplying, subtract when dividing. Write the subtraction out when an index is negative: $6 - (-1) = 7$.
4. **Roots:** root the number, divide the index: $\sqrt{36a^{16}} = 6a^8$.
5. **Check:** positive indices only, and each letter appears once.

Worked examples

Example 1. Simplify $\frac{(2y)^3}{4y^{-2}}$, positive indices.

$$\begin{aligned}\frac{(2y)^3}{4y^{-2}} &= \frac{8y^3}{4y^{-2}} = 2y^{3-(-2)} \\ &= 2y^5\end{aligned}\quad (\text{u})$$

Example 2. Simplify $\sqrt[3]{27b^{12}}$ and $16^{-\frac{3}{4}}$.

$$\begin{aligned}\sqrt[3]{27b^{12}} &= 3b^{\frac{12}{3}} = 3b^4 \\ 16^{-\frac{3}{4}} &= \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \frac{1}{8}\end{aligned}\quad (\text{u})$$

2 Expanding

Two brackets: every term in the first times every term in the second (FOIL). **Three brackets:** expand two of them, then multiply the result by the third, every term by every term. Three terms times two terms gives **six products** before you collect like terms.

Example 3. Expand and simplify $(x+2)(x-1)(x+3)$.

Step 1 $(x+2)(x-1) = x^2 - x + 2x - 2 = x^2 + x - 2$

Step 2 $(x^2 + x - 2)(x+3) = x^3 + 3x^2 + x^2 + 3x - 2x - 6$

Step 3 $= x^3 + 4x^2 + x - 6$ (u)

Check at $x = 1$: the brackets give $3 \times 0 \times 4 = 0$ and $1 + 4 + 1 - 6 = 0$.

Special products (worth knowing by sight): $(a+b)^2 = a^2 + 2ab + b^2$, so $(2x-5)^2 = 4x^2 - 20x + 25$, never $4x^2 + 25$; and $(a-b)(a+b) = a^2 - b^2$.

3 Factorising

Write it in this order (factorising)

- Take out the highest common factor:** the largest number and the lowest power of each letter in every term.
- Count the terms** left in the bracket.
 - Two terms, both squares: **difference of two squares**, $a^2 - b^2 = (a-b)(a+b)$.
 - Three terms, $x^2 + bx + c$: two numbers that **multiply to** c and **add to** b .
 - Three terms, $ax^2 + bx + c$ with $a \neq 1$: multiply $a \times c$, find two numbers that multiply to ac and add to b , **split the middle term**, factorise each pair, take out the common bracket.
- Look again:** can any bracket be factorised further? "Factorise completely" means nothing is left.
- Check:** expand your answer back.

Example 4. Factorise $12a^3b^2 - 18a^2b$.

highest common factor: $6a^2$ (lowest power of a), b (lowest power of b).

$$12a^3b^2 - 18a^2b = 6a^2b(2ab - 3)$$

(u)

Example 6. Factorise $3x^2 + 10x - 8$.

$a \times c = 3 \times (-8) = -24$. Multiply to -24 , add to 10: 12 and -2 .

$$\begin{aligned}3x^2 + 10x - 8 &= 3x^2 + 12x - 2x - 8 \\ &= 3x(x+4) - 2(x+4) \\ &= (x+4)(3x-2)\end{aligned}$$

(u)

Example 5. Factorise $3x^3 - 12x$ completely.

$$3x^3 - 12x = 3x(x^2 - 4) = 3x(x-2)(x+2)$$

When you take out a negative number, **every sign in that bracket changes**: $-2x - 8 = -2(x+4)$.

(u)

4 Algebraic fractions

They follow the rules of number fractions. **Factorise every numerator and denominator first.**

Job	Method
Simplify	Factorise top and bottom, cancel whole common factors (brackets), never single terms.
Multiply	Factorise everything, cancel across, multiply what is left.
Divide	Flip the second fraction and multiply.
Add or subtract	Common denominator = every different factor once. Multiply each numerator by the factors its own denominator is missing. Put the second numerator in brackets when subtracting. Then simplify.

Example 7. Simplify $\frac{2x^2 + 5x - 3}{4x^2 - 1}$.

$$= \frac{(2x - 1)(x + 3)}{(2x - 1)(2x + 1)} = \frac{x + 3}{2x + 1}$$

Example 8. Simplify $\frac{x^2 - 4}{3x} \div \frac{x + 2}{6x^2}$.

$$= \frac{(x - 2)(x + 2)}{3x} \times \frac{6x^2}{x + 2} = 2x(x - 2)$$

Example 9. Write $\frac{4x}{x - 2} - \frac{x + 3}{x + 1}$ as a single fraction.

$$\begin{aligned} &= \frac{4x(x + 1) - (x + 3)(x - 2)}{(x - 2)(x + 1)} \\ &= \frac{4x^2 + 4x - (x^2 + x - 6)}{(x - 2)(x + 1)} \\ &= \frac{3x^2 + 3x + 6}{(x - 2)(x + 1)} = \frac{3(x^2 + x + 2)}{(x - 2)(x + 1)} \end{aligned}$$

(u)

(r)

(u) The bracket makes the -6 become $+6$.

5 Equations with fractions, and changing the subject

Write it in this order (solving)

1. Find the lowest common denominator of **every** term.
2. Multiply **every term on both sides** by it, the whole-number terms too. The fractions disappear.
3. Expand, collect, solve. If an x^2 appears, rearrange to $ax^2 + bx + c = 0$ and factorise.
4. Reject any answer that makes a denominator zero. Check one answer in the original equation.

Example 10. Solve $\frac{3x + 1}{x - 2} - 4 = 0$.

$$\begin{aligned} 3x + 1 - 4(x - 2) &= 0 \\ 3x + 1 - 4x + 8 &= 0 \\ x &= 9 \end{aligned}$$

Example 11. Solve $\frac{6}{x} + \frac{4}{x + 1} = 3$.

Multiply every term by $x(x + 1)$:

$$\begin{aligned} 6(x + 1) + 4x &= 3x(x + 1) \\ 3x^2 - 7x - 6 &= 0 \\ (3x + 2)(x - 3) &= 0 \end{aligned}$$

$$x = 3 \text{ or } x = -\frac{2}{3}$$

(u)

(r)

Example 12. Make x the subject of $y = \frac{2x + 3}{x - 1}$.

When the new subject appears **twice**: collect both terms on one side, then **factorise it out**.

$$\begin{aligned} y(x - 1) &= 2x + 3 \\ xy - y &= 2x + 3 \\ xy - 2x &= y + 3 \\ x(y - 2) &= y + 3 \\ x &= \frac{y + 3}{y - 2} \end{aligned}$$

(r)

6 Versions the marker won't accept

Question	Answer given	Why it falls short
Factorise completely $6x^3y - 15x^2\sqrt{y}$	$3x^2(2xy - 5\sqrt{y})$	Correct but not complete: $\sqrt{y}$ is still common. Markers treat this as Achievement evidence only; $3x^2\sqrt{y}(2x\sqrt{y} - 5)$ is needed for Merit.
Simplify, positive indices	$27y^5$ written as $\frac{27}{y^{-5}}$	A negative index is still showing.
Simplify $\frac{x^2 - x - 12}{4x + 12}$	$\frac{(x - 4)(x + 3)}{4(x + 3)}$	Not simplified: the common bracket has not been cancelled. Markers want the simplified fraction, $\frac{x - 4}{4}$.
Any part	a correct answer followed by a wrong "further simplification"	Stop when the answer is simplest. Extra wrong working can cost the mark.

7 Ways to lose marks

From the wording of the marking schedules and from teaching observation; numbered for marking.

- L1 Negative index left in the answer, or put on the wrong factor.** $3y^{-1} = \frac{3}{y}$, not $\frac{1}{3y}$. "Positive indices" means none negative in the final answer.
- L2 Power or root applied to the letter but not the number.** $(3y)^4 = 81y^4$, not $3y^4$. $\sqrt{36a^{16}} = 6a^8$: root the 36 as well, and halve the index (not root it).
- L3 Terms lost when expanding three brackets.** Two brackets first, then six products. Count them before collecting.
- L4 Not factorised completely.** Take out the full highest common factor, letters and roots included, then check every bracket again.
- L5 Signs not changed when a negative factor is taken out.** $-8x - 20 = -4(2x + 5)$, not $-4(2x - 5)$. Expand the bracket back to check.
- L6 Cancelling terms instead of factors.** In $\frac{x^2 + 3x}{x^2 - 9}$ the x^2 terms do not cancel. Factorise, then cancel brackets.
- L7 Minus sign not carried through the second numerator.** $-(x+3)(x-2) = -(x^2+x-6) = -x^2-x+6$. Bracket first, then change every sign.
- L8 Answer not left in simplest form.** After combining, look for a common factor on top and a bracket that cancels.
- L9 Not every term multiplied by the denominator.** In $\frac{3x+1}{x-2} - 4 = 0$ the 4 is multiplied by $(x-2)$ too.
- L10 Subject left on both sides.** If x is still on the right, it is not the subject. Collect, factorise out x , divide.

8 Quick review

Index laws from Year 11

Multiply	add the indices
Divide	subtract the indices
Power of a power	multiply the indices
Power of a product	every factor, number too
Power zero	equals 1

Number fractions

Simplify	divide top and bottom by the highest common factor
Multiply	tops times tops, bottoms times bottoms
Divide	flip the second, then multiply
Add, subtract	lowest common denominator, then add the tops

$$\frac{5}{6} - \frac{3}{4} = \frac{10}{12} - \frac{9}{12} = \frac{1}{12} \quad (12, \text{not } 24)$$